Did you use a language model?

A : No, I did not find any use for a language model for this assignment.

What did you do to verify that your code worked as expected, whether you used a language model or not?

A : I looked at the sample solution provided in the zipfile.

# Exercise 1-1: Get started with JupyterLab

## Get the data

In [1]:
```
# Imports the Pandas library and stores the dataset into a dataframe
import pandas as pd
poll_url = 'http://projects.fivethirtyeight.com/general-model/president_general_
polls = pd.read_csv(poll_url)
```

In [2]:
```
# Displays the first 5 rows of the data
polls.head()
```

Out[2]:

| | cycle | branch | type | matchup | forecastdate | state | startdate | enddate | |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 2016 | President | polls-plus | Clinton vs. Trump vs. Johnson | 11/8/16 | U.S. | 11/3/2016 | 11/6/2016 | News/Wa |
| **1** | 2016 | President | polls-plus | Clinton vs. Trump vs. Johnson | 11/8/16 | U.S. | 11/1/2016 | 11/7/2016 | Google C |
| **2** | 2016 | President | polls-plus | Clinton vs. Trump vs. Johnson | 11/8/16 | U.S. | 11/2/2016 | 11/6/2016 | |
| **3** | 2016 | President | polls-plus | Clinton vs. Trump vs. Johnson | 11/8/16 | U.S. | 11/4/2016 | 11/7/2016 | |
| **4** | 2016 | President | polls-plus | Clinton vs. Trump vs. Johnson | 11/8/16 | U.S. | 11/3/2016 | 11/6/2016 | Gravis N |

5 rows × 27 columns

In [6]:

```
# Sorts the dataframe according to column name 'state' and displays first 2 rows
polls.sort_values('state', inplace=True)
polls.head(2)
```

Out[6]:

| | cycle | branch | type | matchup | forecastdate | state | startdate | enddate |
|---|---|---|---|---|---|---|---|---|
| **12475** | 2016 | President | polls-only | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 10/21/2016 | 11/2/2016 |
| **4610** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 11/1/2016 | 11/7/2016 |

2 rows × 27 columns

In [18]:

```
# Sort the dataframe acording to the column name 'state' in descending order and
polls.sort_values('state', ascending=False).head()
```

Out[18]:

| | cycle | branch | type | matchup | forecastdate | state | startdate | enddate | |
|---|---|---|---|---|---|---|---|---|---|
| **5595** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Wyoming | 2016-10-24 | 2016-10-30 | Sur |
| **6763** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Wyoming | 2016-09-07 | 2016-09-13 | |
| **5875** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Wyoming | 2016-10-17 | 2016-10-25 | Sur |
| **6892** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Wyoming | 2016-08-31 | 2016-09-06 | |
| **5592** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Wyoming | 2016-10-15 | 2016-10-19 | |

5 rows × 27 columns

# Clean the data

## Keep only the rows with "now-cast" in the type column

In [8]:

```
# Returns the 5 records only where column type is "now-cast"
polls = polls.query('type=="now-cast"')
polls.head()
```

Out[8]:

| | cycle | branch | type | matchup | forecastdate | state | startdate | enddate |
|---|---|---|---|---|---|---|---|---|
| **4610** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 11/1/2016 | 11/7/2016 |
| **5997** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 10/27/2016 | 11/2/2016 |
| **7530** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 8/17/2016 | 8/23/2016 |
| **4369** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 11/1/2016 | 11/7/2016 |
| **5504** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 7/19/2016 | 7/19/2016 |

5 rows × 27 columns

## Rename voting columns

In [9]:

```
# Renaming function to a new name
polls.rename(columns={'rawpoll_clinton':'Clinton_pct',
                      'rawpoll_trump':'Trump_pct'},inplace=True)
polls.head()
```

```
C:\Users\Dell\AppData\Local\Temp\ipykernel_12380\4063675258.py:1: SettingWithCopy
Warning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy
  polls.rename(columns={'rawpoll_clinton':'Clinton_pct',
```

Out[9]:

| | cycle | branch | type | matchup | forecastdate | state | startdate | enddate |
|---|---|---|---|---|---|---|---|---|
| **4610** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 11/1/2016 | 11/7/2016 |
| **5997** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 10/27/2016 | 11/2/2016 |
| **7530** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 8/17/2016 | 8/23/2016 |
| **4369** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 11/1/2016 | 11/7/2016 |
| **5504** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 7/19/2016 | 7/19/2016 |

5 rows × 27 columns

## Convert date columns to datetime objects

In [10]:

```
# Converts 'startdate' and 'enddate' to 'datetime' format using 'pd.to_datetime'
date_cols = ["startdate","enddate"]
polls[date_cols] = polls[date_cols].apply(pd.to_datetime)
polls.head()
```

```
C:\Users\Dell\AppData\Local\Temp\ipykernel_12380\732497925.py:2: SettingWithCopyW
arning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy
  polls[date_cols] = polls[date_cols].apply(pd.to_datetime)
```

Out[10]:

| | cycle | branch | type | matchup | forecastdate | state | startdate | enddate | |
|---|---|---|---|---|---|---|---|---|---|
| **4610** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 2016-11-01 | 2016-11-07 | |
| **5997** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 2016-10-27 | 2016-11-02 | Surv |
| **7530** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 2016-08-17 | 2016-08-23 | |
| **4369** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 2016-11-01 | 2016-11-07 | Surv |
| **5504** | 2016 | President | now-cast | Clinton vs. Trump vs. Johnson | 11/8/16 | Alabama | 2016-07-19 | 2016-07-19 | |

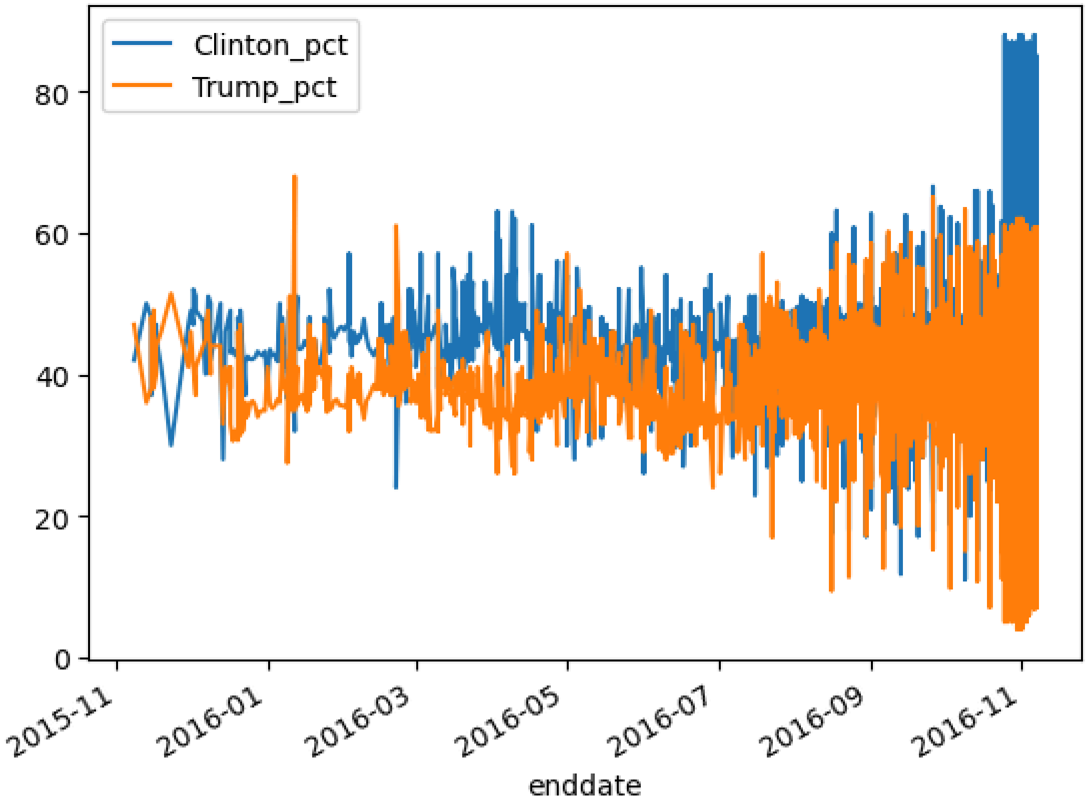
5 rows × 27 columns

## Try two plots

In [11]:

```
# Generates line plot with 'enddate' as x-axis against 'Clinton_pct' and 'Trump_
polls.plot.line(x='enddate',y=['Clinton_pct','Trump_pct'])
```

Out[11]: <Axes: xlabel='enddate'>

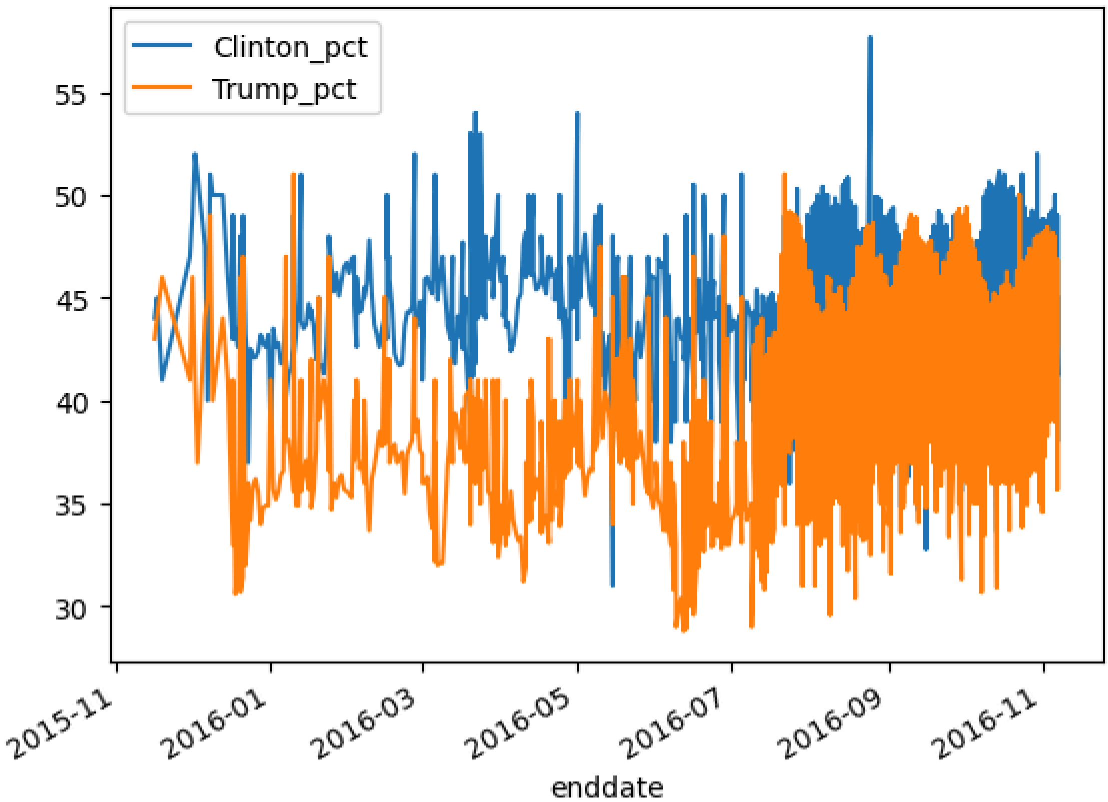



In [14]:
```
%%time
#Magic command to measure execution time of code block

# Plots the line plot post query to get rows where the 'state' is 'U.S'
polls.query('state == "U.S."') \
    .plot(x='enddate',y=['Clinton_pct','Trump_pct'])
```

```
CPU times: total: 15.6 ms
Wall time: 51.9 ms
```

Out[14]: `<Axes: xlabel='enddate'>`




In [15]:
```
# Magic command that shows information about all the variables in the current na

%whos
```

```
Variable    Type         Data/Info
----------------------------------
date_cols   list         n=2
pd          module       <module 'pandas' from 'C:<...>es\\pandas\\__init__.py'>
poll_url    str          http://projects.fivethirt<...>nt_general_polls_2016.csv
polls       DataFrame          cycle      branch   <...>n[4208 rows x 27 columns]
```

# Exploratory Data analysis practice

In [16]:
```
# shows the tyoe of the variable
type(polls)
```

Out[16]:
```
pandas.core.frame.DataFrame
```

In [17]:
```
#
polls.describe()
```

Out[17]:

| | cycle | startdate | enddate | samplesize | poll_wt | Clinto |
|---|---|---|---|---|---|---|
| **count** | 4208.0 | 4208 | 4208 | 4207.000000 | 4208.000000 | 4208.0( |
| **mean** | 2016.0 | 2016-08-31 01:50:11.406844160 | 2016-09-06 14:23:02.509505792 | 1148.216068 | 0.255590 | 41.9! |
| **min** | 2016.0 | 2015-11-06 00:00:00 | 2015-11-08 00:00:00 | 35.000000 | 0.000000 | 11.0 |
| **25%** | 2016.0 | 2016-08-10 00:00:00 | 2016-08-21 00:00:00 | 447.500000 | 0.000377 | 38.0( |
| **50%** | 2016.0 | 2016-09-23 00:00:00 | 2016-09-30 00:00:00 | 772.000000 | 0.008317 | 43.0( |
| **75%** | 2016.0 | 2016-10-20 00:00:00 | 2016-10-28 00:00:00 | 1236.500000 | 0.087304 | 46.2( |
| **max** | 2016.0 | 2016-11-06 00:00:00 | 2016-11-07 00:00:00 | 84292.000000 | 8.720654 | 88.0( |
| **std** | 0.0 | NaN | NaN | 2630.856265 | 0.669328 | 7.7 |

In [19]:

```
# Will display information such as data type of columns
polls.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 4208 entries, 4610 to 5595
Data columns (total 27 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   cycle              4208 non-null   int64
 1   branch             4208 non-null   object
 2   type               4208 non-null   object
 3   matchup            4208 non-null   object
 4   forecastdate       4208 non-null   object
 5   state              4208 non-null   object
 6   startdate          4208 non-null   datetime64[ns]
 7   enddate            4208 non-null   datetime64[ns]
 8   pollster           4208 non-null   object
 9   grade              3779 non-null   object
 10  samplesize         4207 non-null   float64
 11  population         4208 non-null   object
 12  poll_wt            4208 non-null   float64
 13  Clinton_pct        4208 non-null   float64
 14  Trump_pct          4208 non-null   float64
 15  rawpoll_johnson    2799 non-null   float64
 16  rawpoll_mcmullin   30 non-null     float64
 17  adjpoll_clinton    4208 non-null   float64
 18  adjpoll_trump      4208 non-null   float64
 19  adjpoll_johnson    2799 non-null   float64
 20  adjpoll_mcmullin   30 non-null     float64
 21  multiversions      12 non-null     object
 22  url                4207 non-null   object
 23  poll_id            4208 non-null   int64
 24  question_id        4208 non-null   int64
 25  createddate        4208 non-null   object
 26  timestamp          4208 non-null   object
dtypes: datetime64[ns](2), float64(10), int64(3), object(12)
memory usage: 920.5+ KB
```

In [20]:

```
# Provides number of rows and columns
polls.shape
```

Out[20]:

```
(4208, 27)
```

In [ ]: